TUBULAR CONFINEMENT AND CLOCKED CHEBYSHEV CANCELLATION

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1. Definitions

Let

$$\rho_k = \beta_k + i\gamma_k \tag{1.1}$$

denote a nontrivial zero of the Riemann zeta function, ordered by increasing ordinate γ_k.

Definition 1.1 (Logarithmic clock)**.** For $k \geq 3$, define

$$\tau_k = \pi \frac{\log\log k}{\log k}. \tag{1.2}$$

Definition 1.2 (Free-amplitude phase-walk)**.** For $\sigma \in \mathbb{R}$, define

$$z_k(\sigma) = \exp\big((\sigma + i\gamma_k)\tau_k\big), \tag{1.3}$$

and

$$S_N(\sigma) = \sum_{k=3}^{N} z_k(\sigma). \tag{1.4}$$

Definition 1.3 (Clocked Chebyshev scale)**.** Define

$$x_k = \exp(\tau_k). \tag{1.5}$$

Equivalently,

$$\log x_k = \tau_k = \pi \frac{\log\log k}{\log k}. \tag{1.6}$$

Definition 1.4 (Tubular confinement)**.** The walk $S_N(\sigma)$ is said to be tubarly confined if there exist $C \in \mathbb{C}$, $R > 0$, and $N_0 \geq 3$ such that

$$|S_N(\sigma) - C| \leq R \tag{1.7}$$

for every $N \geq N_0$.

Definition 1.5 (Clocked Chebyshev cancellation)**.** The clocked Chebyshev terms are said to have controlled cancellation at amplitude σ if there exist $C \in \mathbb{C}$, $R > 0$, and $N_0 \geq 3$ such that

$$\left| \sum_{k=3}^{N} x_k^{\sigma} \left(\cos(\gamma_k \log x_k) + i \sin(\gamma_k \log x_k)\right) - C \right| \leq R \tag{1.8}$$

for every $N \geq N_0$.

2. Equivalence theorem

Theorem 2.1 (Tubular confinement is clocked Chebyshev cancellation)**.** *For every $\sigma \in \mathbb{R}$, the following statements are equivalent.*

(i) *The walk $S_N(\sigma)$ is tubarly confined:*

$$|S_N(\sigma) - C| \leq R \tag{2.1}$$

for every $N \geq N_0$.

(ii) *The clocked Chebyshev terms have controlled cancellation:*

$$\left| \sum_{k=3}^{N} x_k^{\sigma} \left(\cos(\gamma_k \log x_k) + i \sin(\gamma_k \log x_k) \right) - C \right| \leq R \tag{2.2}$$

for every $N \geq N_0$.

Proof. By (1.5) and (1.6), one has

$$x_k^{\sigma} = \exp(\sigma \tau_k), \tag{2.3}$$

and

$$\gamma_k \log x_k = \gamma_k \tau_k. \tag{2.4}$$

Therefore

$$x_k^{\sigma} \left(\cos(\gamma_k \log x_k) + i \sin(\gamma_k \log x_k) \right) = \exp(\sigma \tau_k) \left(\cos(\gamma_k \tau_k) + i \sin(\gamma_k \tau_k) \right) \tag{2.5}$$

$$= \exp(\sigma \tau_k) \exp(i \gamma_k \tau_k) \tag{2.6}$$

$$= \exp\left((\sigma + i\gamma_k) \tau_k \right) \tag{2.7}$$

$$= z_k(\sigma). \tag{2.8}$$

Summing (2.8) from $k = 3$ to N gives

$$\sum_{k=3}^{N} x_k^{\sigma} \left(\cos(\gamma_k \log x_k) + i \sin(\gamma_k \log x_k) \right) = S_N(\sigma). \tag{2.9}$$

Assume first that $S_N(\sigma)$ is tubarly confined. Then (2.1) holds. By the identity (2.9), this is exactly (2.2). Thus tubular confinement implies controlled clocked Chebyshev cancellation.

Conversely, assume that controlled clocked Chebyshev cancellation holds. Then (2.2) holds. By (2.9), the partial sum appearing in (2.2) is exactly $S_N(\sigma)$. Therefore (2.1) holds. Thus controlled clocked Chebyshev cancellation implies tubular confinement.

The two statements are equivalent. □

3. Critical amplitude

Proposition 3.1 (Critical amplitude)**.** *At*

$$\sigma = \frac{1}{2}, \tag{3.1}$$

the clocked Chebyshev amplitude is

$$x_k^{\sigma} = x_k^{1/2}. \tag{3.2}$$

Thus $\sigma = 1/2$ is the critical amplitude level of the clocked Chebyshev model.

Proof. Substitution of (3.1) into (2.3) gives

$$x_k^{\sigma} = x_k^{1/2}. \tag{3.3}$$

For a zero $\rho = \beta + i\gamma$, the standard zero term has the algebraic form

$$x^{\rho} = x^{\beta}\left(\cos(\gamma \log x) + i \sin(\gamma \log x)\right). \tag{3.4}$$

Hence the exponent $\beta = 1/2$ corresponds to the amplitude $x^{1/2}$. In the clocked model, the free parameter σ plays the same formal amplitude role, and $\sigma = 1/2$ is the critical value. □

4. Conclusion

The exact identity

$$S_N(\sigma) = \sum_{k=3}^{N} x_k^{\sigma}\left(\cos(\gamma_k \log x_k) + i \sin(\gamma_k \log x_k)\right) \tag{4.1}$$

with

$$\log x_k = \pi \frac{\log\log k}{\log k} \tag{4.2}$$

shows that the logarithmic phase-walk is not merely analogous to a Chebyshev-type oscillatory sum. It is exactly a Chebyshev-shaped oscillatory sum evaluated on the clocked scale (4.2).

Therefore,

(4.3)

$$\boxed{\text{tubular confinement of } S_N(\sigma) \Longleftrightarrow \text{controlled cancellation of clocked Chebyshev-type terms}}$$

for every real σ.

The critical case is

$$\boxed{\sigma = \frac{1}{2},} \tag{4.4}$$

because this is exactly the value for which the clocked amplitude is $x_k^{1/2}$.